

Physics 0525: Lab 3

AC Response of RC and RLC Circuits

Equipment: Function generator, DMM, oscilloscope, resistors, capacitors, inductors

Part 1: RC low-pass filter

Build the RC low-pass filter shown on Figure 1. Measure and record the actual values of the components and use those for calculations! The input comes from the signal generator. Measure the output across the capacitor with the oscilloscope.

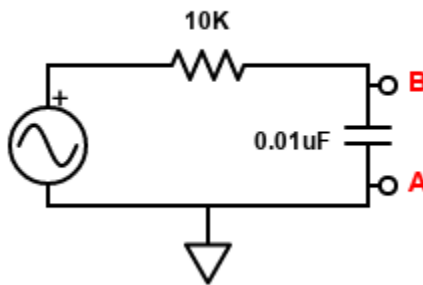


Figure 1: RC low-pass filter.

You will measure the filter response $A(f) = V_{out}/V_{in}$ as a function of frequency, and compare it to the theoretical formula, where V_{in} and V_{out} represents the amplitude of the input and output sine wave:

$$\frac{V_{out}}{V_{in}} = \frac{1}{(1 + \omega^2 R^2 C^2)^{1/2}}$$

$$\omega = 2\pi f$$

$R = 9.9 \text{ k}\Omega$

$C = 10.2 \text{ nF}$

(1 V_{pp})

* Analysis: Calculate the -3dB cut-off frequency.

$$\frac{V_{out}}{V_{in}} = \frac{-3}{20 \log_{10} \frac{V_{out}}{V_{in}}}$$

$$\hookrightarrow \frac{V_{out}}{V_{in}} = \frac{20 \log_{10}}{-3} = \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}} \Rightarrow \sqrt{2} = \sqrt{1 + \omega^2 R^2 C^2}$$

$$\hookrightarrow 2 = 1 + \omega^2 R^2 C^2 \Rightarrow R^2 C^2 = \omega^2, f_c = \frac{2\pi}{(9.9 \times 10^3)(10.2 \times 10^{-9})} =$$

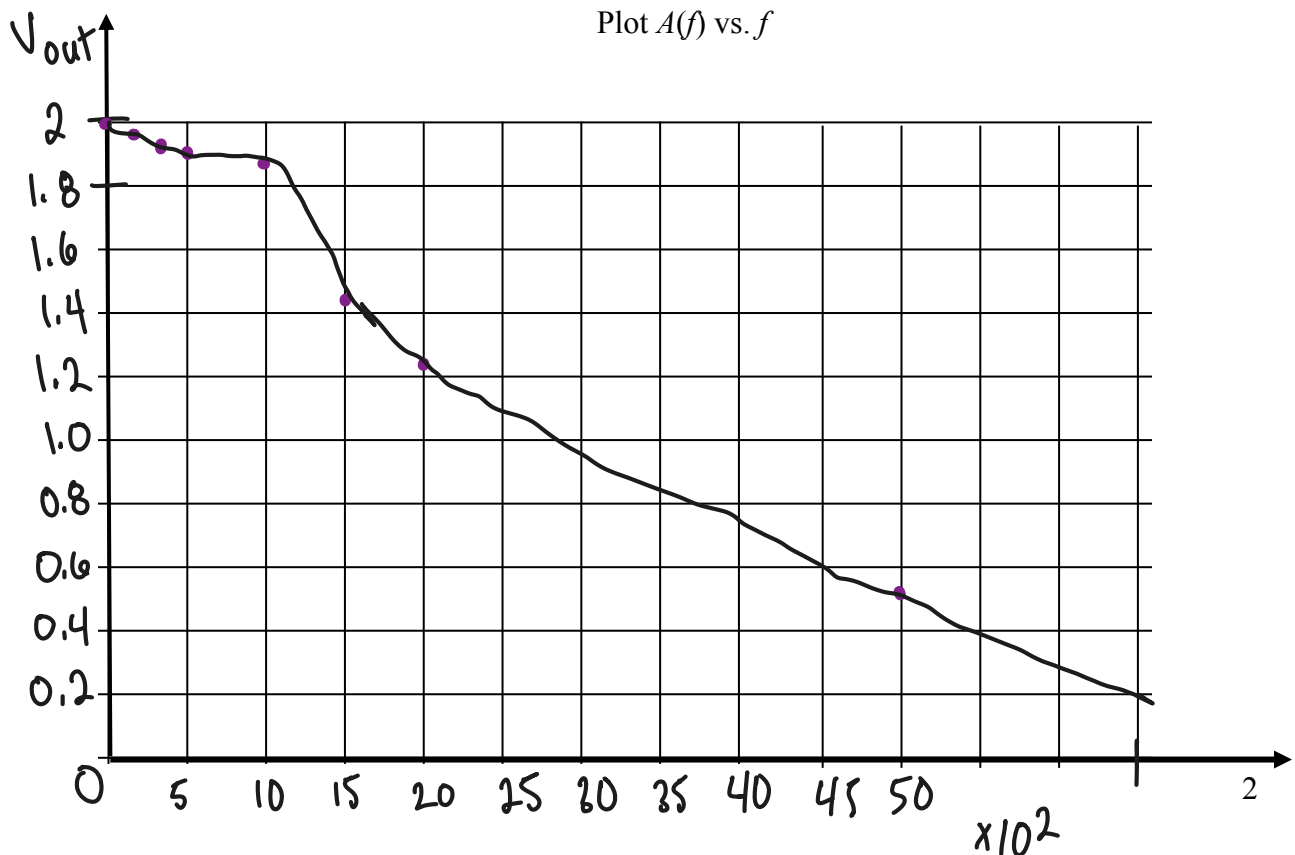
Expected -3dB cut-off frequency: $f_c = \underline{1563} \text{ [Hz]}$

Name _____

Date _____

Use a sine wave with amplitude 1 Vpp (or more) from the signal generator as input. Display both input and output on the oscilloscope simultaneously. Measure the amplitude of input and output at a series of frequencies and fill in the following table. To get the amplitude use the Measurement function of the scope. Make sure at least 2-3 full cycles of the waveforms are visible, and the top and bottom of the waveforms are within the vertical range of the scope otherwise the measurement will be incorrect! Select a wide range of f values starting far below the cut-off frequency and going far above it.

f [Hz]	ω [s^{-1}]	V_{in}	V_{out}	V_{out}/V_{in} (exp)	V_{out}/V_{in} (theory)
1	2π	2	1.96	$1.96/2 =$	
10	20π	1.96	1.94	$1.94/1.96 =$	
50	100π	1.992	1.976	$1.976/1.992 =$	
75	150π	1.992	1.972		
100	200π	1.992	1.96		
200	400π	1.992	1.952		
500	$10,000\pi$	1.976	1.880		
1000	$2E3\pi$	↓	1.864		
1500	$3E3\pi$		1.424		
1563.			1.392		
2000	$4E3\pi$		1.216		
5000	$1E4\pi$		0.584		
10,000	$2E4\pi$		.31		
15,000	$3E4\pi$		.206		
20,000	$4E4\pi$		.156		
30,000	$6E4\pi$		.106		



Indicate the -3dB cut-off frequency on the graph.

How does the amplitude of the output change with frequency? Why is this circuit called a low-pass filter?

the ROC of the amplitude increases dramatically between 1000 Hz and 1500 Hz and the output voltage drop levels off, becoming more and more linear as the frequency increases past the cutoff frequency.

Next, observe the input and output simultaneously on the oscilloscope as you change the frequency from far below the -3 dB cutoff frequency to far above the cutoff frequency.

What happens to the phase difference between input and output as you vary f , and why?

Be specific, describe which leads or lags the other (Hint: which one reaches its maximum first?)

The output voltage lags behind the input as we increase the frequency. This occurs due to the nature of the RC circuits' filter response as a result of input f .

Measure the phase difference at the -3dB cut-off frequency.

$$\Delta t = 82 \mu s$$

$$\text{Phase} = \frac{\Delta t}{\text{Period}} = \frac{82 \times 10^{-6} s}{637.551 \times 10^{-6} s} = 0.12867 \text{ rad/s} (360^\circ) = 46.3^\circ$$

Part 2: RC high-pass filter

Change the circuit to the configuration of an RC high-pass filter. Use the same circuit components as in part 1.

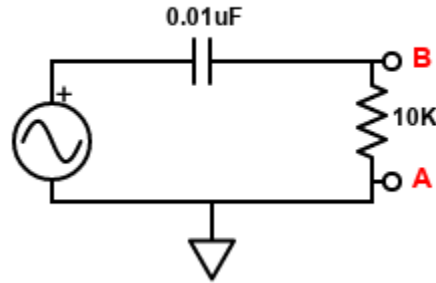


Figure 2: RC high-pass filter

The response function is now given by: $\frac{V_{out}}{V_{in}} = \frac{R}{(R^2 + 1/\omega^2 C^2)^{1/2}}$

Analysis: Calculate the -3dB cut-off frequency.

$$\frac{1}{\sqrt{2}} = \frac{R}{(R^2 + \frac{1}{\omega^2 C^2})^{1/2}} \Rightarrow \frac{1}{R\sqrt{2}} = \left(R^2 + \frac{1}{\omega^2 C^2}\right)^{1/2} \Rightarrow R\sqrt{2} = \left(R^2 + \frac{1}{\omega^2 C^2}\right)^{1/2}$$

$$\hookrightarrow 2R^2 = R^2 + \frac{1}{\omega^2 C^2} \Rightarrow \frac{1}{R^2 C^2} = \omega^2$$

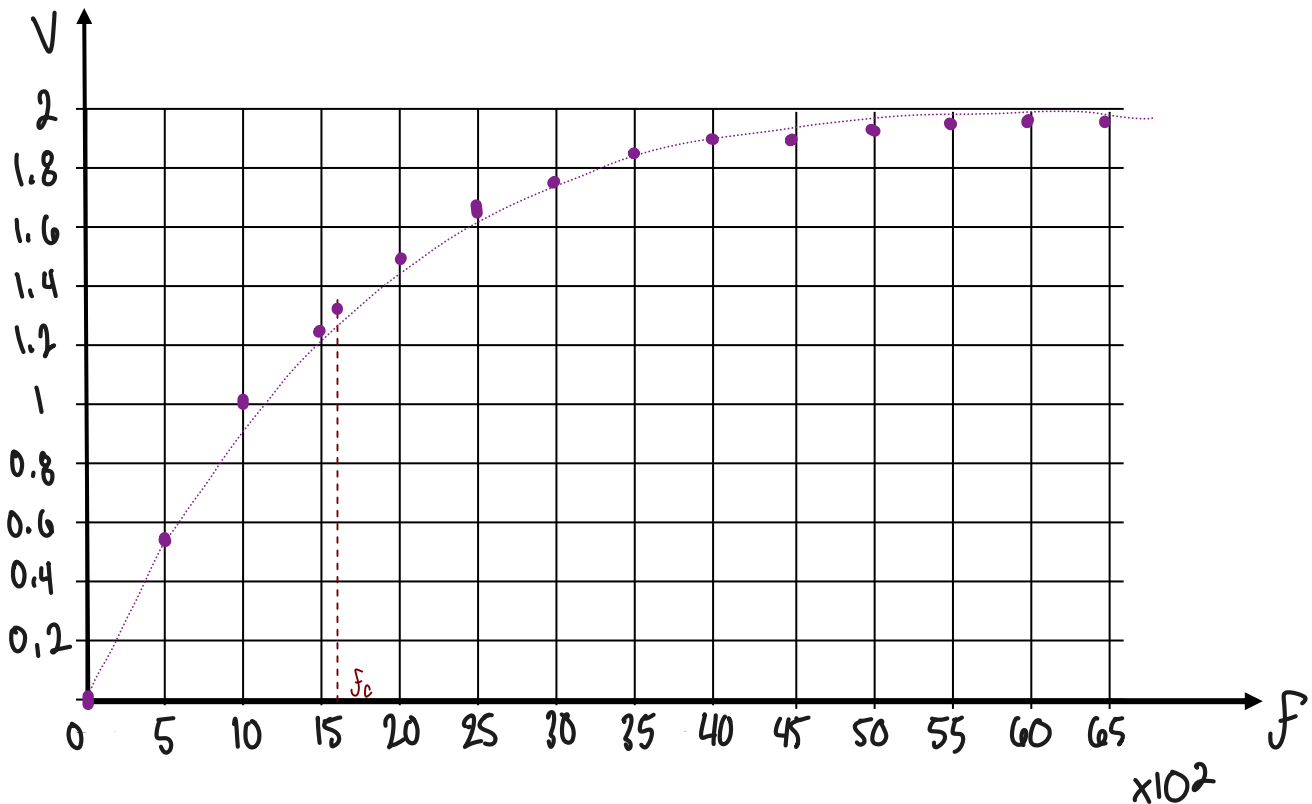
Expected -3dB cut-off frequency: $f_c = \underline{1572.5}$ [Hz]

Name _____

Date _____

Use a sine wave from the signal generator as input and display input and output simultaneously on the oscilloscope. Measure the amplitudes at a series of frequencies and fill in the following table. Be sure to choose wide range of f values!

f [Hz]	ω [s^{-1}]	V_{in}	V_{out}	V_{out}/V_{in} (exp)	V_{out}/V_{in} (theory)
1	2π	1.977	0		
500		1.992	0.588		
1000			1.04		
1500			1.344		
1577.5			1.376		
2000			1.536		
2500			1.648		
3000			1.728		
3500			1.726		
4000			1.824		
4500			1.84		
5000			.864		
5500			.872		
6000			.888		
6500			.904		
7000			.909		

Plot $A(f)$ vs. f 

Name _____

Date _____

How does the amplitude of the output change with frequency? Why is this circuit called a high-pass filter?

The output amplitude linearly rises until reaching the cutoff frequency, where the rate of change approaches 0.

Next, observe input and output simultaneously on the oscilloscope as you change the frequency from far below the -3 dB cutoff frequency to far above the -3 dB cutoff frequency. What happens to the phase difference between input and output as you vary f , and why? Be specific, describe which leads or lags the other, and by what phase.

The input lags behind the output, increasing in phase difference as the frequency increases and switches above.

Measure the phase difference at the -3 dB cut-off frequency?

$$\Delta t = \frac{79.2 \mu s}{\text{Period}} = \frac{79.2 \times 10^{-6}}{634.477 \times 10^{-6}} = 0.1248 (360^\circ) \approx 45^\circ$$

Part 3: LC resonance (band-pass filter)

A parallel circuit consisting of an inductor and a capacitor will resonate at the frequency

$$f_{res} = \frac{1}{2\pi\sqrt{LC}}$$

Build the circuit in Figure 3. Don't forget to measure and use the actual values of your components.

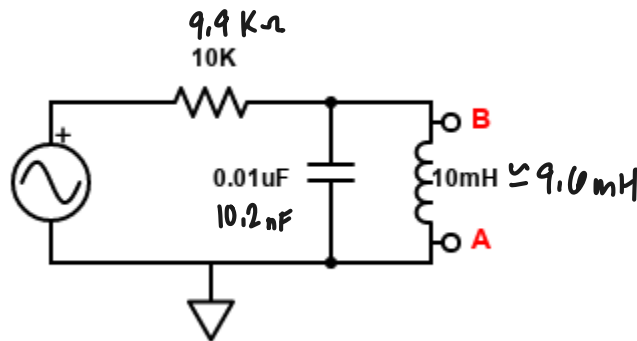


Figure 3: LC resonance

Calculate the resonant frequency of the circuit.

$$f_c = \frac{1}{2\pi\sqrt{LC}} \Rightarrow \frac{1}{2\pi\sqrt{(9.9 \times 10^{-3})(10.2 \times 10^{-9})}} \Rightarrow \frac{1}{6.275 \times 10^{-5}} = 1.6083 \times 10^4$$

Expected resonant frequency: $f_{res} = \underline{1.6083635 \times 10^4}$ [Hz]

Measure the resonant frequency by applying sine waves with frequencies around the calculated resonant frequency and searching for maximum output amplitude.

Measured resonant frequency: $f_{res} = \underline{16275}$ [Hz]

Name _____

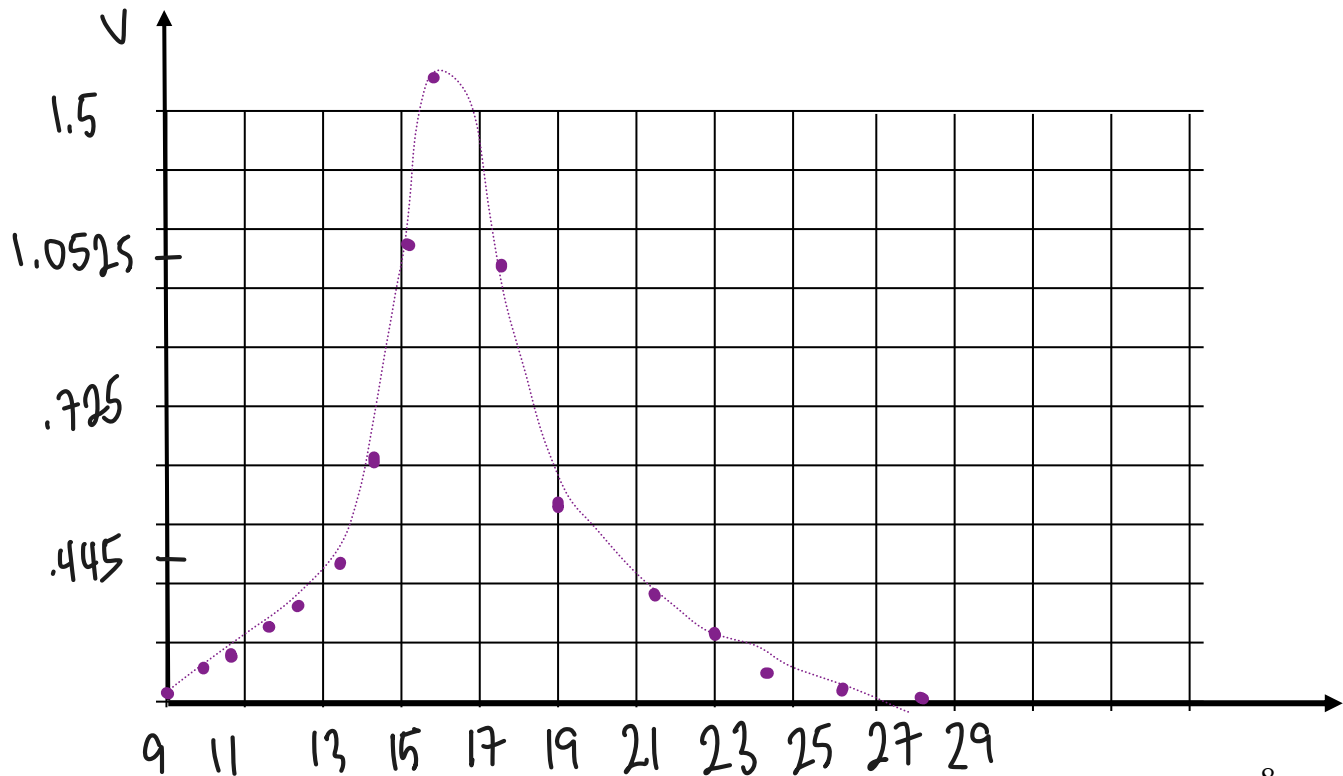
Date _____

Use the oscilloscope to display and measure the amplitude of input and output at a series of frequencies. Scan the frequency over a range for which the output rises from about 10% of its maximum and then falls off again to about 10% of the maximum.

Data Table:

f [Hz]	ω [s^{-1}]	V_{in}	V_{out}	V_{out}/V_{in} (exp)	V_{out}/V_{in} (theory)
9275		1.976	0.16		
10275			0.2		
10775 *			0.216		
11275			0.248		
12775 *			0.396		
13275			0.44		
14275			0.656		
15275			1.088		
16275			1.584		
17775 *			0.908		
* 19275			0.536		
* 21275			0.344		
22275			0.288		
23275			0.256		
24275			0.232		
* 26275			0.192		
* 28275			0.160		

* discontinuities *

Plot $A(f)$ vs. f 

Name _____

Date _____

How does the amplitude of the output change with frequency? Why is this circuit called a band-pass filter?

the amplitude of the output is filtered through a narrow band around the resonance frequency, where the amplitude increases towards it, growing exp. as it approaches and by symm. dropping

Next, observe input and output simultaneously on the oscilloscope as you change the frequency from far below f_{res} to far above f_{res} . What happens to the phase difference between input and output as you vary f , and why? Be specific, describe which leads or lags the other. *when past,*

Below resonance frequency the output lags behind the input signal, approaching phase then after fully in phase the output signal begins to lag again.

What is the phase difference at the resonance frequency?

0

Name _____

Date _____

What is the amplitude of the input sine wave?

$$V_{in} = \underline{1.984} \text{ [V]}$$

What is the maximum amplitude of the output sine wave at the resonance frequency?

$$V_{max} = \underline{1.584} \text{ [V]}$$

Question: You notice that the maximum amplitude of the output is smaller than the amplitude of the input signal. That is because the parasitic resistance of the real capacitor and inductor will lead to a loss. The “equivalent circuit” at the resonance frequency is shown in Figure 4. Calculate the resistance R_{loss} from V_{in} and V_{max} .

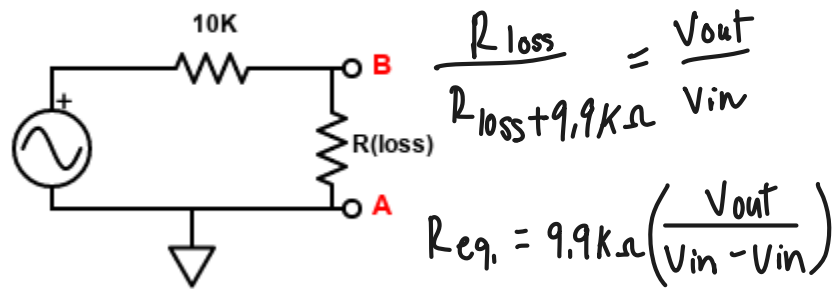


Figure 4: LC equivalent circuit at resonance. $R_{loss} \approx 392 \text{ k}\Omega$

$$R_{loss} = \underline{39006} \text{ } [\Omega]$$